

Detection and Alerting Animals in Forest using Artificial Intelligence and IoT

Girish H
Associate Professor
Dept. of ECE
Cambridge Insitute of
Technology
Bengaluru, India
hgirishphd@gmail.com

Manjunath T G
Associate Professor
Dept. of EEE
Sai Vidya Institute of
Technology
Bengaluru, India
tgmtnath@gmail.com

Vikramathithan A C
Associate Professor
Dept. of ECE
Sai Vidya Institute of
Technology
Bengaluru, India
vikramathithanac@gmail.com

Abstract— One significant issue that every country confronting today is passing and wounds because of street mishaps. The human-creature crash is one of the significant explanations for some lethal mishaps. There are road-accidents happening like clockwork all throughout the planet. This paper proposes a system to detect animals and alert vehicles to decrease thruway mishaps of animals. Using IOT, a detection and alert system is proposed. This alert system further produces signals in such a way to alert the drivers in the highway if the detection is triggered. IoT plays a major role along with Pi module to detect the animals and alert the vehicles. Sensors are used to detect the obstacle which activates Pi Camera configured by Raspberry Pi to capture the live images or video and movements of animals with help of image detection then alert the people and vehicles in the forest highways. An indication with a light at particular distance during night time is also incorporated. Thus, the paper aim is to secure animals from accidents and save many lives from danger. YOLO (You just look once) algorithm is proposed which assists with handling 45 frames each second and takes the entire picture to anticipate the object.

Keywords— *Internet of Things, Raspberry Pi, YOLO,*

I. INTRODUCTION

The problem many countries are facing nowadays is death, injuries and accidents in the roads. The human-animal collision is one of the major reasons behind many fatal accidents. There are road accidents happening every 15 minutes around the globe. The conflict between human-beings and animals is that leads to damage of crops, death of peoples and people kills animals. Especially rare species of animals are killed in the forest highways very frequently. To avoid animal-vehicle collisions, these days we have been using various sign boards to warn people but still we are facing many accidents on the road, this may be due to the negligence of drivers, or their inability to notice the sign boards, or many times their ignorance as they exactly cannot predict if they are going to encounter any wild animal or not. The Apps are being developed for detecting animals play important role is to provide methods to various problems is real-life. The main cause for many applications are the detection of wild-animals in the image/video. So, we decided to come up with an idea to protect animals from the accident and to reduce the number of kills happening around the globe. The purpose of animal detection is to prevent the number of animal-vehicle collisions and aim is to design a system which can be installed on highways at certain distance from the road or to be installed on vehicles. And

efficiently detect animals with high accuracy to alert the vehicles to slow down speed.

A detection system and accident avoidance system by using CV Technique and Technologies used are a histogram of oriented gradients (HOG), Cascade classier and OpenCV libraries has been proposed [1]. A strategy used for detecting the space between the animal from the camera location in true units.

The wild-animal detecting and colliding avoidance system by system by using Deep Learning has designed [2]. The Input is of images were utilized from Open Image Dataset, COCO Dataset, and PASCAL VOC Dataset. The animal detection system by using deep learning has discussed [3],[4] and also Convolutional Neural Network, Robotically Recognizing, Description of Animals in Camera Images, and Counting with Deep-Learning, Template Matching Algorithm were considered. This paper discusses about the usage of the Convolutional Neural Network (CNN) algorithm mainly used to identify the animals in the wild.

The search term is animal detection using machine learning. This paper discusses about the usage of the CNN algorithm mainly used to identify the animals in the wild. This algorithm has a very high efficiency in detecting the animals displayed with a very high accuracy.

The animal detecting system by using is CNN has been conferred [5]. The technologies is used are Data collection, Identification network and improvements. Model training and Model validation. This paper discusses about the usage of the Convolutional Neural Network (CNN) algorithm identifies the animals in the wild. The authors designed animal detection system by using DCNN model [6]-[8]. The cameras are shaft mounted along with boundaries of the ranch and PIR Sensors. The monitoring system and detection using IoT was discussed [9],[10].

Dissimilar Meta Heuristics Schemes in Intelligent Fault Detection and reconfiguration were conversed [11]-[3]. Jungle Fire Threatening Arrangement by means of drones and Artificial Intelligence [14]and also Centered on GIS and WSNs techniques are used [15],[16]. The prevailing topic of wireless sensor grids as a possible result to the competitive of primary detection of forest fervors has been emphasized[17]. The device offered varieties of sensors attached, solar recharging appliance, and wireless facts transmission to realize the task. The scheme is executed exploiting sensors, controller board, Wi-fi and GPS in conjunction with proper programming to device the traffic stream[18].

II. INTERNET OF THINGS

The IoT is a technique comprises of connecting devices along with manmade digital machines, objects, animals or people. The IoT devices have a discrete identification and can pass without person-person or person-PC interaction. In the next few years, technologies based on the Internet of Things will provide advanced services and almost change people's daily life. Innovations in medicine field, energy, irrigation and smart city/homes, gene therapy are typical examples of solid establishment of the Internet of Things. As of now, more than 9 billion "things" have been connected to the Internet. Within a span of time, number is expected to reach a staggering 20 billion.

IoT uses mainly four components:

A. Low-power embedded system

In the design of Electronic systems, high performance and low battery power consumption are the opposite factors that play an important role.

B. Cloud Computing-

The processing information collection from Internet of Things equipment is massive and needs to be saved on a trusted storing server. The cloud computing plays very important role for this. Once the information is processed and understood, it provides more space to find electrical faults or errors and other problems in the system.

C. Big data availability-

Internet of Things is highly dependent on sensors which capture real time values. The electronic equipment is spread out widely across various fields, its use will release massive streams of big data.

D. Network Connection:

Internet connection is essential for communication, that every physical object is denoted by an Internet Protocol address. Henceforth, depending on the IP name, the number of addresses available is limited. As the number of devices continues to increase, this representation of the system will no longer be viable. Hence, the innovators are thinking for other alternative denoting system to depict every physical object.

III. DEEP LEARNING

Deep learning is a subdivision of machine learning in the branch of artificial intelligence, with networks that can learn from unstructured data teachers or data without labelling is also known as deep neural learning or deep neuronal network. Deep learning is a simple automatic learning, which is potentially inspired by the structure of the human brain.

The characteristics of focus on the work of deep learning algorithms should try to draw the same conclusions by continuously analysing the data with human being that receive a given logical structure. To achieve this function, deep learning uses a multilayer structure of an algorithm called neuronal network.

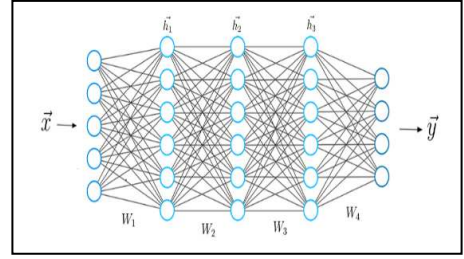


Fig. 1. Structure of Neural Network

Fig 1 depicts the structure of neural network, which comprises of an input-layer and an output-layer. Each input layer has a single input and a single output corresponding the input.

The basic unit of the neural network is known as a 'neuron', there are input neurons, output neurons corresponding to the input and output layers respectively, additionally there are hidden neurons in the network, and are present in the hidden layer whose main functionality is to apply weights to the input and direct them through an activation function as output. They perform non-linear transformations of the inputs and direct them.

IV. SOFTWARE REQUIREMENTS

A. Tensor flow

You only see it once is an object detection algorithm. Compared with the recognition algorithm, the detection algorithm can not only predict the category label, but also detect the location of the object. So it can not only classify images into one category, but also detect multiple objects in an image.

And this algorithm is not based on multiple neural networks. Apply a one of the network in neuron to the complete picture. The neural network splits the picture into multiple regions and predicts the binding boxes and probable combination for all regions. These binding boxes are weighing by the predicted probability.

Raspbian OS:

Raspberry-Pi Operating System is a open source Debian-based operating system, which is optimized for hardware based Raspberry-Pi. The Raspberry-Pi OS is available with 35,000 software packages, and the software is pre-compiled included in a good format for installing easily on the Raspberry Pi and its OS is compatible for the Raspberry Pi series of optimized one-board PCs with ARM processors. Raspberry-Pi could be run on desktop platform, with an open frame stackable window manager and unique themes.

Raspberry Pi OS has 3 installation options:

- Raspberry Pi Operating System Lite is the tiny version and does not have the desktop platform.
- The Raspberry-Pi operating system comprises of the Pixel desktop platform.
- Raspberry-Pi Operating System is available as pre installable with cumulative productivity software.

B. ReactJS

React is an precise, flexible & declarative Java Script library for creating user interfaces. It is the "V" in MVC. ReactJS is an proprietary front end library which is responsible for the visual layer of the application which is vindicated by Facebook.

C. Google Maps API

The Google Maps API is a efficient software platform that can be used to create custom maps, search maps, log functions, display location-synced real-time data, plan routes, and more. Basically, it creates an agile and personalized experience to bring the real world to its customers with static and dynamic maps and images 360° views.

D. MongoDB

MongoDB (MDB) is a proprietary NoSQL DB. NoSQL is an complementary to traditional Data bases. NoSQL DB are very useful for working with large distributed data sets. MongoDB is a software platform that allows you to maintain document-oriented data, store and recover it. MongoDB.

V. HARDWARE REQUIREMENTS

A. Raspberry- Pi 3 Model B

The raspberry-pi 3 is the generation 3 raspberry-pi. This efficient small sized PCB where PC could be used for various appliances and has many features than the pi model B+. Fig 2 show the hardware of Raspberry Pi 3 Model B.



Fig. 2. Raspberry Pi 3 Model B

The raspberry pi 3 model B comes with a more efficient processor. It is 10x speed than the 1st generation raspberry-pi and also it includes wireless LAN and has Bluetooth connections which makes it the perfect method for various products.

B. Pi Camera

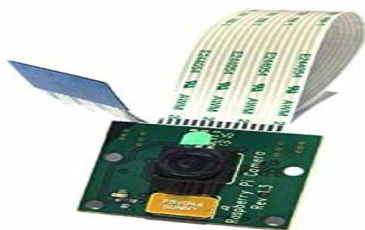


Fig. 3. Pi-Camera

The Pi-camera is a small sized, very light weighted camera which can be used for Raspberry pi. The camera module is shown in Fig 3. The camera interacts with Raspberry pi uses the protocol based MIPI camera serial interface which is commonly used in processing of picture, ML techniques and also in theft detection projects. Other than this, Raspberry pi also can use normal USB webcams which are utilized alongside PC.

VI. METHODOLOGY AND SYSTEM ARCHITECTURE

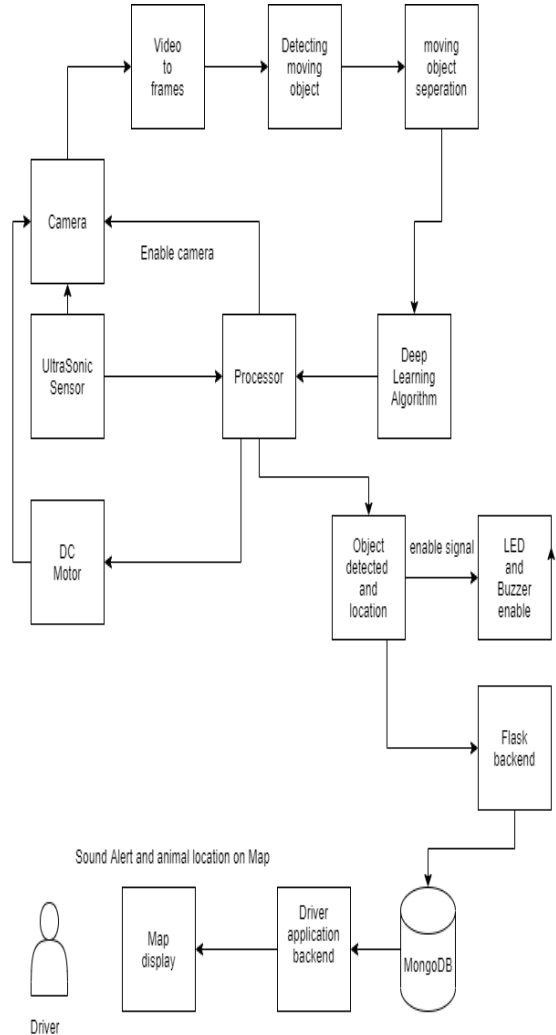


Fig. 4. System Architecture

The efficiency of system would be highly dependent on the capacity of the ultrasonic sensors to detect the objects in the ambience accurately. Fig 4 depicts the system architecture of deep learning based animal detection and alerting system. The Camera would be triggered to capture the video when the sensor senses an animal. The captured video is converted into frames and sent to the model which is trained on animal dataset using YOLO algorithm which is the most efficient and accurate image detection algorithm.

If the model detects an animal in the images, it will classify it and also output the class to which it belongs to (it detects which animal is present in the image), and the animal class and location is stored in the database using the flask backend. A signal is sent to enable the LED lights and buzzer as an alert system. The driver alert application backend queries the

database for any new entry at that time and send it to the frontend that is a Map display.

The driver is alerted through sound notification and shows the location of the animal on the map. The driver will be notified to slow down to avoid any collision with the animal. This system can efficiently reduce the number of human animal collisions and thereby prevent road accidents caused due to wild animal encounter.

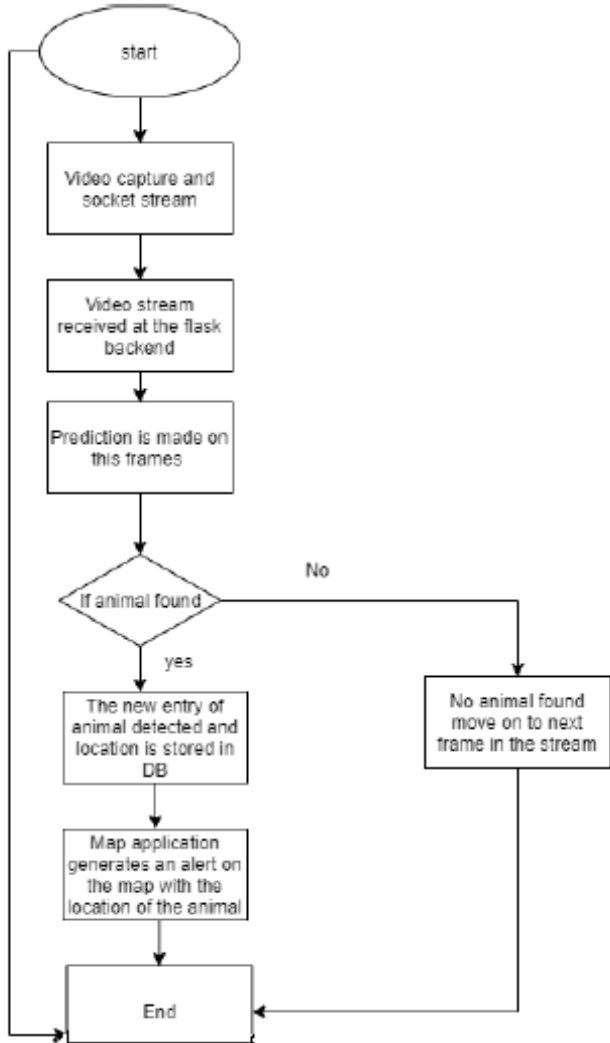


Fig. 5. Flow Chart of YOLO algorithm

YOLO mean you only look once is a things detecting algorithm for real time applications. Because of competition, it is not a commonly used classifier designed as an things detector. Fig 5 shows the flow chart of YOLO algorithm.

YOLO working principle is to divide the given image to a small grid of $S \times S$ cells, in which every cell is used for the prediction of 5 bounding boxes describes the object with a rectangle. This generates a definite point, that is a measurable than that of a thing is locked.

Henceforth, the score has nothing to do along its type of objects present in the box, only the shape of the box. The predicted binding box, it will predict a class and operates as a regular classifier, giving the probable combination of all possible classes. The bounding box confidence score and the category prediction are combined into a end point that specifies the probable method that every box contains a

particular type of object. In view of these design options, every box has low confidence points, so only boxes so that its end score exceeds the threshold are retained.

YOLO detects the model as a reversion issue. Therefore, the given picture is divided into an $S \times S$ grid, and for each cell in the grid, the bounding box B , the confidence of these boxes, and the class C probability are detected. These encoded detection are as an $.S \times S \times (B * 5 + C)$. Fig 6 shows the Tester for grid.

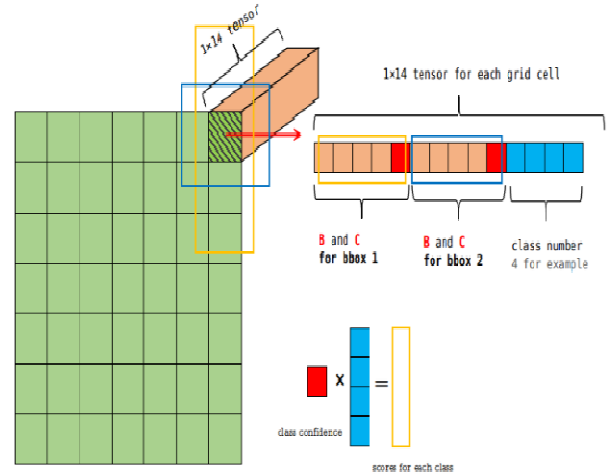


Fig. 6. Tester for grid

VII. RESULTS AND DISCUSSION

In this paper the raspberry pi is gadget is utilized to distinguish creatures with assistance of Pi camera and ultrasonic sensors. Here the yield is to identify creatures and caution the vehicles in preceding stay away from the quantity of kills on the wood's interstates. Sound buzzer, furthermore, LED Light and the Driver Alert application are the two significant part in giving the alarm to the vehicles. The DC Motor is utilized for rotational reason.

The entire arrangement will be put together in the wood's roadways at a specific spot and distance. The ultrasonic sensor will detect the item when the creature is close by to the road and afterward the camera will be powered once the sensor gives the caution. The camera will capture the live images of the animals and then sends the images to deep learning model which detects the animal class. The buzzer and the LED are turned on to alert any vehicle around the location.

The class and the location of the animal is stored in the database. This information is used by the driver alert application which keeps scanning the database for any new entries every second. The Map displays gives a sound notification, and the location of the animal is marked on the map.

Now every one of the vehicles will hinder the speed for the creatures to cross or to cruise by. This outcome in a smaller number of mishaps and harm to the creatures just as the vehicles on highways. Fig 7 illustrates object detection of class elephant. Fig 8 spectacles the object detection of class tiger and Fig 9 shows the driver alert application map.



Fig. 7. Object detection of Class Elephant

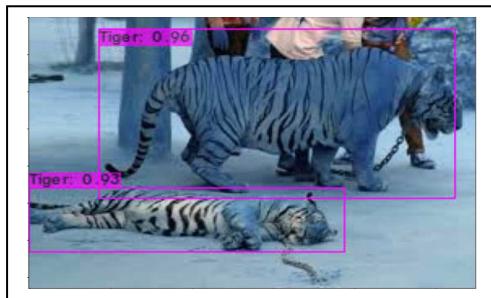


Fig. 8. Object detection of class Tiger

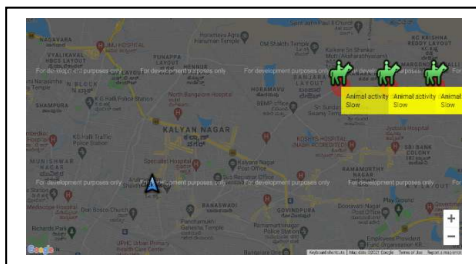


Fig. 9. Driver Alert Application Map

VIII.CONCLUSION

Today we are in the realm of machines practically the world is running under deep learning methods. We have been utilizing various sorts of machines in our everyday life utilizing numerous calculations. In this paper an attempt was made for animal detection for wildlife using IoT and Deep learning. An attempt was made in this paper is to identify creatures and ready vehicles in forest highways. There are numerous creatures got harmed in woodland interstate mishaps this generally brings about animal's passing's, by this rare species are likewise getting influenced. Raspberry pi 3 model B is used to distinguish the creatures and to caution the vehicle.

The ultrasonic sensor is utilized to distinguish the deterrent once the hindrance recognized, the pi camera start empowers, and it will catch live pictures of the creatures and the picture is analysed with the information base utilizing profound learning calculation. Then, at that point it will ready vehicle in the wood's thruways.

Future scopes integrating the system in the vehicles to detect in long range and autonomously reducing the speed of the vehicle to avoid collision.

REFERENCES

- [1] Sharma, S. U., & Shah, D. J. (2016).. IEEE Access, 5, 347-358.
- [2] Saxena, A., Gupta, D. K., & Singh, S. (2021).. In Advances in Communication and Computational Technology (pp. 1069-1084). Springer, Singapore.
- [3] Harish, S., Rao, S., & Chethan, P. (2019, July).. In 2019 1st International Conference on Advances in Information Technology (ICAIT) (pp. 540-543). IEEE.
- [4] Banupriya Prakash, Animal Det. using Deep Learning Algorith., First International Conference on Intelligent Digital Transformation ICIDT - 2019 (11-13 July 2019, Volume - I)
- [5] Divya, U. K., & Praveen, M. (2018). Iot-based wild animal intrusion det. scheme. International Journal on Recent and Innovation Trends in Computing and Communication, 6(7).
- [6] Hefeeda, M. and Bagheri, M. Forest Fire Modeling and Early Detection. Ad Hoc & Sensor Wireless Networks , 7, 169-224, 2009.
- [7] Zhang, J., Li, W., Han, N. and Kan, J. Forest Fire Det. System Based on a ZigBee WSN. Frontiers of Forestry in China , 3, 369-374,2008.
- [8] Wang, M., Liu, H., Chen, F. and Liu, J. Forest Fire Warning System Based on GIS and WSNs. 4th International Conference on Advanced Information Technology and Sensor Application, 21-23 August 2015,Harbin, 3-6,2015.
- [9] Girish H, Shashikumar D R, "Emission monitoring system using IOT", Dogo Rangsang Research Journal, UGC Care Group I Journal, ISSN: 347-7180, Vol – 8 Issue-14, No.6, 2020.
- [10] Girish H, Sridhar S, Bharath H, Vishvesh V, Gowtham K V, "IOT based transformer power theft detection and protection", IJER, Vol-5, ISSN: 2347-5013, may 2016Writer's Handbook. Mill Valley, CA: University Science, 1989.
- [11] T.G.Manjunath,Ashok Kusagur "Analysis of Different Meta Heuristics Method in Intelligent Fault Detection of Multilevel Inverter with Photo voltaic Power Generation Source",2018,DOI: <http://doi.org/10.11591/ijpeds.v9.i3>
- [12] T.G. Manjunath, Ashok Kusagur. "Robust Fault Detection of Multilevel Inverter using Optimized Radial Basis Function based Artificial Neural Network in Renewable Energy Power Generation Application" 180(48):8-15, June 2018. DOI -10.5120/ijca2018917231.
- [13] T.G. Manjunath, Ashok Kusagur "Fault Diagnosis and Reconfigure of Multi level Inverter switch failure- A performance perspective" .Vol.6, No.6, December 2016, pp.2610-2620 ISSN:2088-8708,DOI:10.11591/ijece.v6i6.12112.
- [14] Cuesta-Frau, D., Araujo, A., Alejandre, J. and Rozas, A. Forest Monitoring and Wildland Early Fire Detection by a Hierarchical WSN. Journal of Sensors , 2016, Article ID:8325845,2016
- [15] Wang, M., Liu, H., Chen, F. and Liu, J. Forest Fire Warning System Based on GIS and WSNs. 4th International Conference on Advanced Information Technology and Sensor Application, 21-23 August 2015,Harbin, 3-6,2015.
- [16] Chauhan, A., Semwal, S. and Chawhan, R. ANN-Based Forest Fire Detection System Using Wireless Sensor Network. India Conference ,Mumbai,13-15,December2013,1-6.<https://doi.org/10.1109/INDCON.2013.6725913>.
- [17] Owayjan, M., Freiha, G., Achkar, R., Abdo, E. and Mallah, S. Firoxio: Forest Fire Detection and Alerting System. 17th IEEE Mediterranean Electro mechanical Conference , Beirut, 13-16 April 2014,177-181, 2014.
- [18] Girish H. "Intelligent Traffic Tracking System using Wi-Fi ." International Journal for Scientific Research and Development 8.12 (2021): 86-90